Residual-corrected closed-loop action

Original marked scalar; P-192 primary; five-curve exact census

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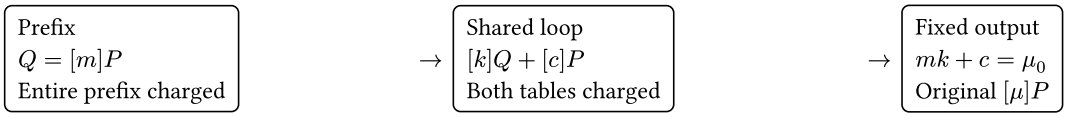
Executable point action

Keep the original μ and write its centered representative as $\mu_0 = mk + c$. At relation radix $m = \ell$, the checked closure supplies $Q = \sigma\eta\varphi(P) = [\ell]P$, and the evaluator computes

$$[\mu]P = [k]Q + [c]P.$$

Both signed-digit terms share the main doubling loop. The complete fused prefix, both tables, additions and negations are charged. The separate explicit-map/endpoint checks return the same points. The global noninteger quotient remains unexpanded; the implementation asserts the displayed rational-subgroup action.

Point computation; the output scalar stays fixed



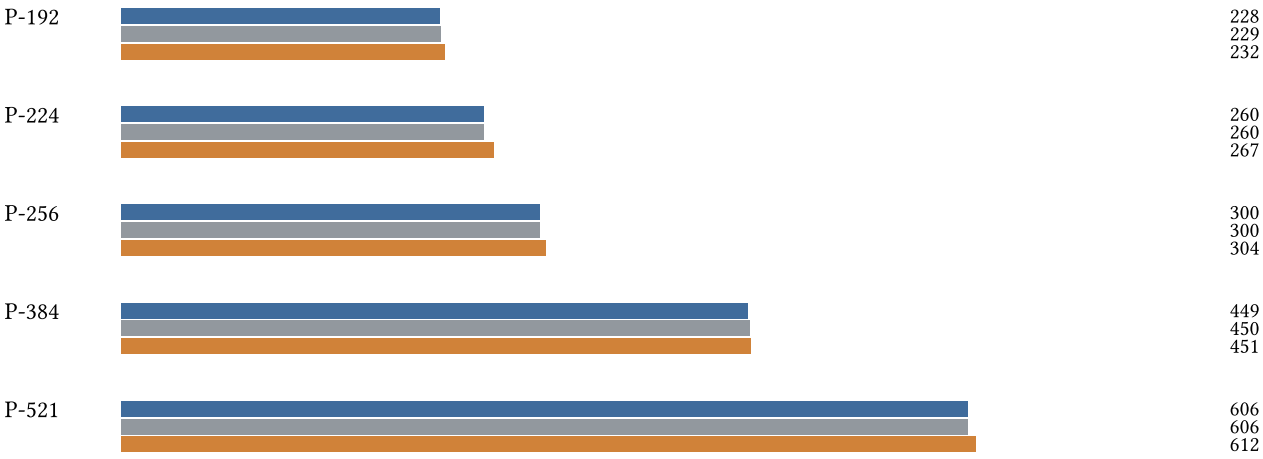
Frozen primary comparison

Curve	Direct	Binary split	Relation split
P-192	228	229	232
P-224	260	260	267
P-256	300	300	304
P-384	449	450	451
P-521	606	606	612

Values are minimum unweighted D+A over the declared finite design. The relation is above both controls on every curve, including each declared weight2..4 sensitivity. This is the preregistered comparison; no runtime inference follows.

Exact D+A minima for the original fixed scalar

Blue: direct. Grey: binary split control. Orange: relation residual split.



One complete frozen42,480-row census. All raw schedules and weights2..4 retained. D+A counts explicit group API calls; no clock or population inference.

Construction and qualification

For each radix, the centered Euclidean remainder r fixes $c = r + mv$, $k = \frac{\mu_0 - c}{m}$. Every integer offset $-64 \le v \le 64$ and window2..7 is retained. Binary controls use the identical joint evaluator at radices1,2,4,8,16,32. The direct reference repeats all centered-scalar shifts −256..256 and the same six windows in this instrument. No scalar, generator, subgroup or point changes across arms.

All42,480 schedules completed once. Independent GMP replay reconstructs every row and20minimum sets without including the BIGNUM producer. All300 selected-plan point/count checks pass. Earlier software qualification passes3150 point/count cases,450 full explicit-map/closing-isomorphism cases and five wrong-remainder controls. Five post-scan corruptions reject: wrong selected plan, wrong raw counter with updated checksum, missing row, duplicate row and changed protocol. Fresh archive replay is byte-identical; deterministic repacking matches.

Complete selected relation vectors

Curve	D	A	N	Table entries
P-192	192	40	19	6
P-224	222	45	19	10
P-256	252	52	24	10
P-384	379	72	26	18
P-521	521	91	33	18

For historical engineering context, the earlier translated zero-remainder recipe selected235,273,309,458,613 D+A. Its design differs; the decisive controls are those in the current table. P-192′s new vector removes four doubles, adds one addition and two negations, and removes three table entries against that earlier recipe. The remaining direct-reference gap is four D+A calls on P-192 and two on P-384. Retaining useful prefix intermediates is the next source-level opportunity, with identical reuse required for binary controls.

Publication, resources and scope

Source/checker commit df8001b87da1 and freeze publication206ba5c3e9bd precede data. Protocol body SHA256: 31b28e2913c78fc1c78c819f8190af17b777d4f18b810dec9120cc22071fe335 Both binaries/sources, seven input hashes, finite grids and tie rule are bound. The canonical export validates; its60 cases are derived minimum records, not60 independent runs. Raw archive104,052bytes, SHA256: f45bf33a443157615b1a7a68992fe14db5ce3fc1e10a65e85a37e7bc671e6b70

Initial compiler indentation warnings, the first stopping-rule schema error and an incorrect export-schema lookup remain documented. No scientific scan was repeated. Each invocation was capped at120seconds, one process/thread and2GiB. No new cloud job, controller, role, cap, budget or generator was changed. Existing-host billing was not measured. No cryptographic time was collected; the full cold objective still requires plan preparation, allocations, normalization, conversion, verification and destruction on an admitted host.

Construction and source review: ../../RESIDUAL_SPLIT.md. Replay/raw custody: ../../residual-v1/README.md. Standard simultaneous-scalar background: Grabner–Heuberger–Prodinger, helmut-prodinger.at/pdffiles/joint.pdf, introduction/section2; read primary author PDF. Independent window expansions are used, with no JSF optimality or novelty claim. KB unavailable; migrated wrapper absent, exit127. Double-base author PDF returned502 and is not relied on.